



Unit : 1

Consumer Equilibrium

Chapter 2 | Lecture 2



20 utils

Total Utility = 50 utils

①



20 utils

②



+ 16 utils

③



+ 10 utils

④



+ 4 utils

TOTAL UTILITY (TU)

Total utility refers to the total satisfaction obtained from the consumption of all possible units of a commodity. It measures the total satisfaction obtained from consumption of all the units of that good.

TU can be calculated as:

$$TU_n = U_1 + U_2 + U_3 + \dots + U_n$$

Where:

TU_n = Total utility from n units of a given commodity

$U_1 + U_2 + U_3 + \dots + U_n$ = Utility from the 1st, 2nd, 3rd nth unit

n = Number of units consumed

Do Consider it

1. The utility derived from the first unit of a commodity is known as initial utility. For example: Utility of 20 utils from the first ice-cream is termed as the initial utility.
2. Total utility is zero at zero level of consumption.

2 ice cream
TU = 36 utils

(3 ice cream)
TU = 46 utils

2 ice cream | 3 ice cream
TU = 36 utils | TU = 46 utils

+1 = Utility + ?

$46 - 36 = 10 \text{ utils}$

∴ Marginal
Utility

10 utils

MU



1



2



3

Marginal utility

+1

Utility? ↑

$$MU = \frac{\Delta \text{ in } TU}{\Delta \text{ in Units}} = \text{New} - \text{old}$$
$$\frac{46 - 36}{3 - 2} = \frac{TU_3 - TU_2}{3 \text{ units} - 2 \text{ units}}$$
$$= \frac{10}{1}$$

MU = 10 utils

$$3 \text{ ice cream} = TU_3 = 46 \text{ utils (New)}$$

$$2 \text{ ice cream} = TU_2 = 36 \text{ utils (old)}$$

$$MU = TU_n - TU_{n-1}$$

$$= TU_3 - TU_2$$

$$= 46 - 36$$

$$MU = 10 \text{ utils}$$

MARGINAL UTILITY (MU)

Marginal utility is the additional utility derived from the consumption of one more unit of the given commodity. It is the utility derived from the last unit of a commodity purchased. In the words of Chapman, "Marginal utility is addition made to total utility by consuming one more unit of a commodity".

MU can be calculated as $MU_n = TU_n - TU_{n-1}$

Where: MU_n = Marginal utility from nth unit; TU = Total utility from n units;

TU_{n-1} = Total utility from n - 1 units; n = Number of units of consumption

One More way to Calculate MU

MU is the change in TU when one more unit is consumed. However, when change in units consumed is more than one, then MU can also be calculated as:

$$MU = \frac{\text{Change in Total Utility}}{\text{Change in number of units}} = \frac{\Delta TU}{\Delta Q}$$

Total Utility is Summation of Marginal Utilities

Total utility can also be calculated as the sum of marginal utilities from all units,

i.e.

$$TU_n = MU_1 + MU_2 + MU_3 + \dots + MU_n \text{ or simply,}$$

$$TU = \sum MU$$

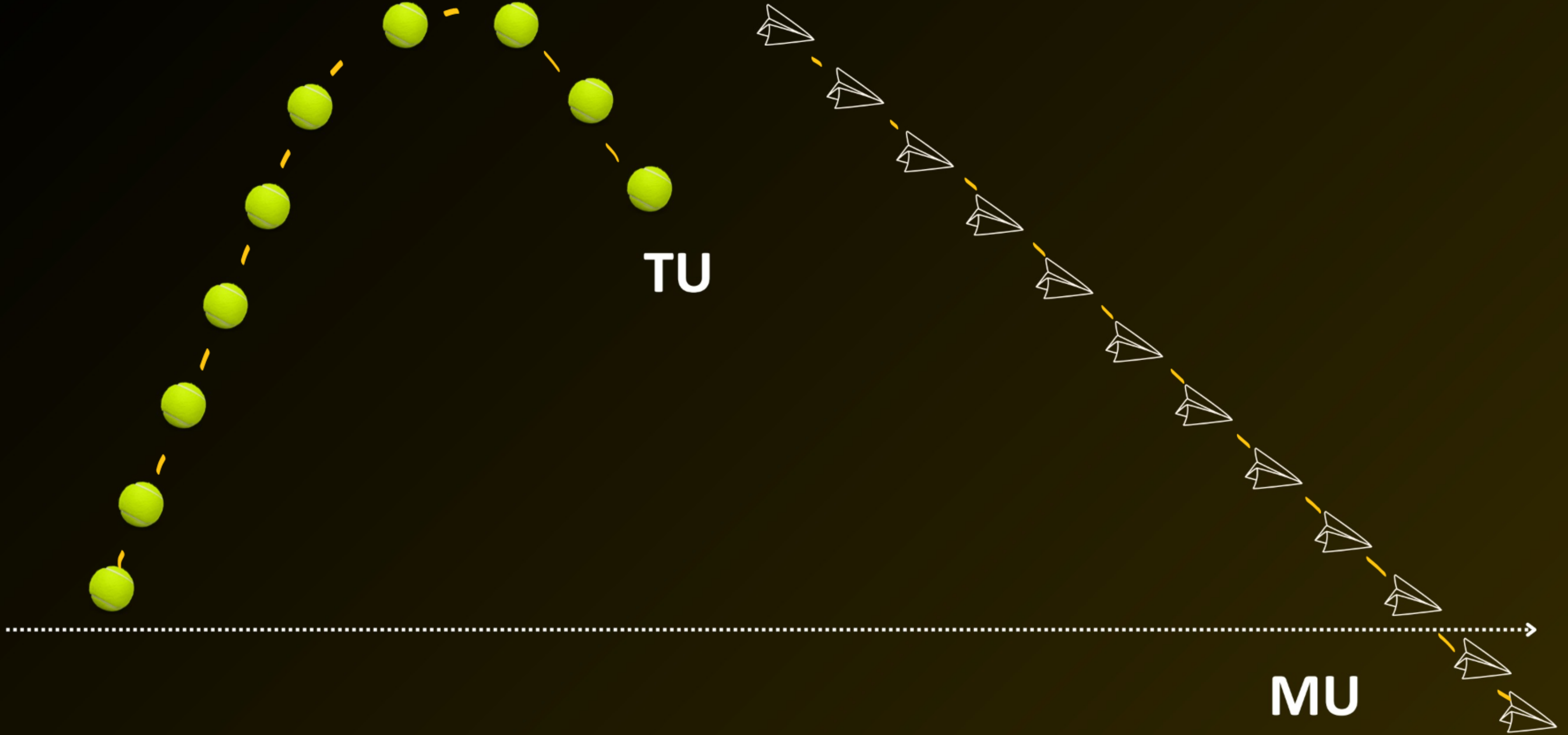
The concepts of TU and MU can be better understood from the following diagram:

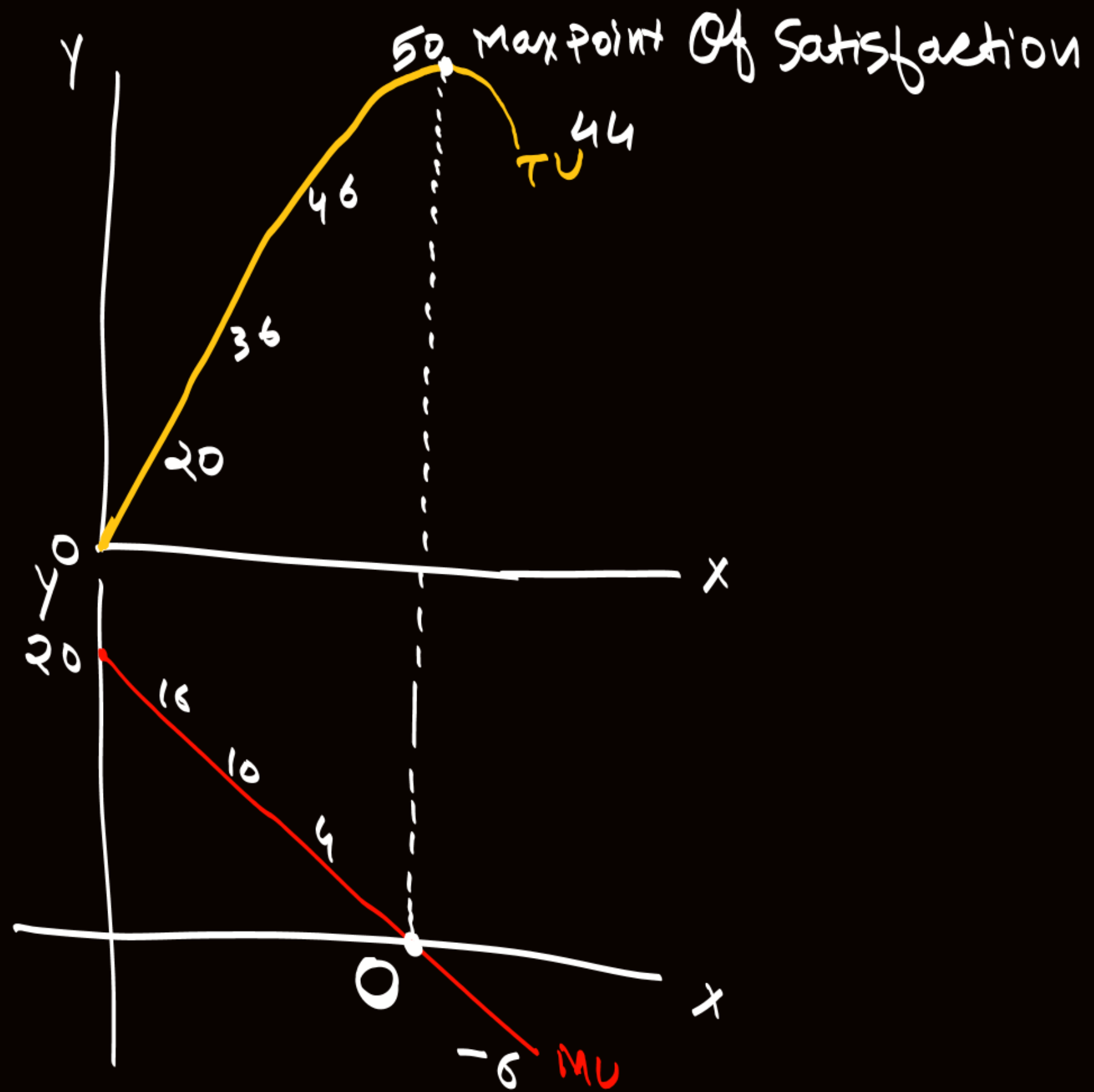
Ice-creams Consumed	Marginal Utility (MU)	Total Utility (TU)
1	20	20
2	16	$20 + 16 = 36$
3	10	$20 + 16 + 10 = 46$
4	4	$20 + 16 + 10 + 4 = 50$
5	0	$20 + 16 + 10 + 4 + 0 = 50$
6	-6	$20 + 16 + 10 + 4 + 0 + (-6) = 44$



EXPLORE MORE ABOUT MARGINAL UTILITY

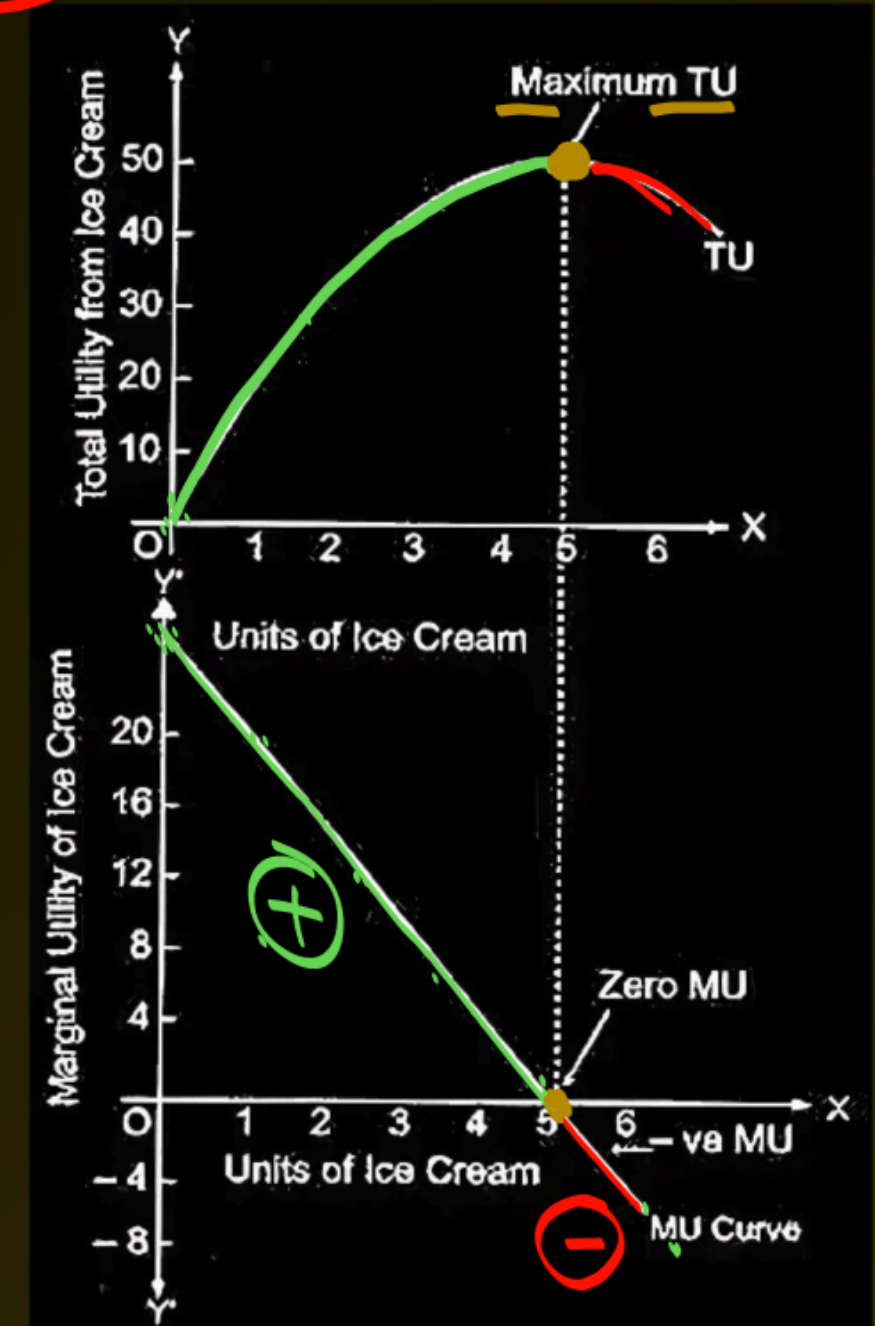
- **Positive Marginal Utility**: If total utility increases from consumption of additional units of a commodity, then marginal utilities of these units will be positive.
- **Zero Marginal Utility**: If the consumption of an additional unit of a commodity causes no change in the total utility, then marginal utility of the additional unit is zero. At this level of consumption, total utility is at its maximum. This point is also known as the point of satiety or the point of maximum satisfaction.
- **Negative Marginal Utility**: If the consumption of an additional unit of a commodity causes a fall in the total utility, it means that the marginal utility of that unit is negative. Negative utility is also known as disutility.





RELATION BETWEEN TU & MU

1. TU starts from origin / increases at diminishing rate / reaches top / starts falling.
2. MU starts from top / falling (DMU) / becomes 0 / then negative.
3. When TU is rising / MU is falling but positive.
4. When TU is at top / MU is 0.
5. When TU is falling / MU is negative.

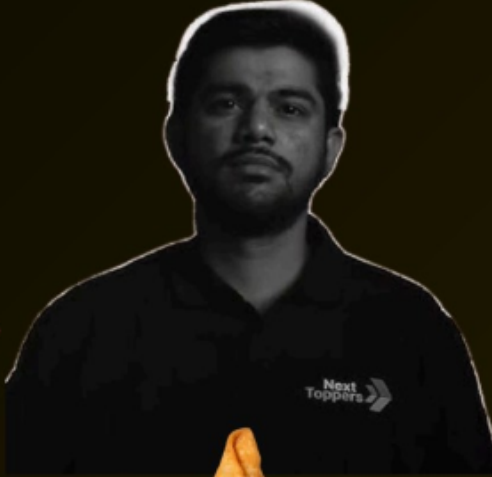




16
1



10
2



4
3



0
4



-6

LAW OF DIMINISHING MARGINAL UTILITY

- Law of diminishing marginal utility (DMU) states that as we consume more and more units of a commodity, the utility derived from each successive unit goes on decreasing.
- People prefer a variety of goods because consuming more and more of any one good reduces the marginal satisfaction derived from further consumption of the same good.
- This law expresses an important relationship between utility and the quantity consumed of a commodity.
- Law of DMU has universal applicability and applies to all goods and services
- This law was first given by a German Economist H.H. Gossen. That is why, it is also known as 'Gossen's first law of consumption'.

**THANK
YOU**

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